Discovering Big Five Personality Profiles through UMAP and HDBSCAN Clustering

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Abstract

This project aims to decode Big Five personality profiles from spontaneous social media language. Analyzing status updates from the "myPersonality" Facebook dataset, we employ Sentence-BERT (`all-MiniLM-L6-v2`) to generate semantic text embeddings, project them using Uniform Manifold Approximation and Projection (UMAP), and perform density-based partitioning using HDBSCAN. To avoid user-activity bias, personality scores are standardized using a population baseline fitted on unique users. Our pipeline identifies 32 distinct clusters, capturing 56.2% of the dataset. A cluster-level TF-IDF analysis reveals clear semantic cohorts ranging from movies to academic discussions. By comparing standardized cluster means against the global population baseline, we show that online linguistic behaviors map to distinct, sometimes counter-intuitive, Big Five profiles. These findings highlight how social media serves as a complex medium for emotional expression, moving beyond typical self-report behavioral assumptions.

Research Question: To what extent can distinct types of social media content be identified through density-based clustering, and how do these linguistic clusters map to the Big Five (OCEAN) personality profiles of their authors?

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1 Introduction

Social media language provides a continuous, valid alternative to traditional self-report questionnaires prone to cognitive and social desirability biases. This project aims to map spontaneous status updates to Big Five personality profiles using unsupervised semantic clustering to understand how everyday online language reflects underlying emotional expression and personality dynamics. To overcome unstructured text challenges like varying density and high noise, we implement a pipeline combining Sentence-BERT (all-MiniLM-L6-v2), UMAP, and HDBSCAN. After generating SBERT embeddings, UMAP reduces the feature space to 5 dimensions, allowing HDBSCAN to isolate distinct semantic cohorts without predefining cluster counts. Crucially, personality scores are standardized via a `StandardScaler` fitted strictly on unique users to ensure an unbiased baseline. Ultimately, our analysis reveals that spontaneous online communication reflects complex emotional coping, interpersonal venting, and social broadcasting, presenting a dynamic view of personality expression that challenges conventional psychological assumptions.

2 Dataset

2.1 Dataset Description

To implement this project, we rely on the "myPersonality" dataset. The myPersonality project was a Facebook application that allowed users to participate in psychological research by completing psychometric personality questionnaires in exchange for feedback on their scores. Approximately 40% of the participating users consented to donate their Facebook status updates data for research purposes.

In this dataset, an individual's personality is mapped using the Big Five framework, which includes:

- **Openness**: The extent to which a person is imaginative, curious, and open to new experiences.
- **Extraversion**: The degree to which an individual is outgoing, and energetic.
- **Agreeableness**: Reflects how compassionate, cooperative, and trusting a person is.
- **Conscientiousness**: The degree of self-discipline, organization, and goal-directed behavior.
- **Neuroticism**: Emotional instability and the tendency to experience negative emotions such as anxiety.

Although the distribution of the official dataset was discontinued in 2018, a small, public subset remains accessible for academic research on GitHub, which serves as the data source for this project.[2]

2.2 Data Profiling

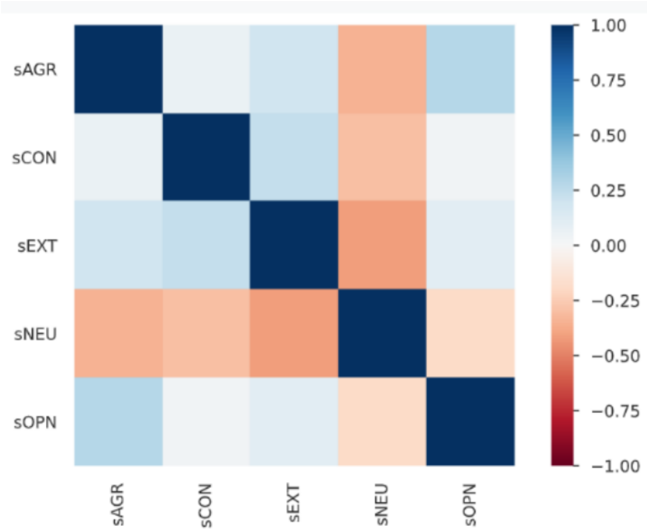

Figure 1: Correlation matrix of Big Five personality variables.

By leveraging the `ProfileReport` function, we generated a comprehensive exploratory report of the data. The raw dataset contains 9,917 rows and 20 columns. For our analysis, we retained seven essential columns:

- `AUTHID`: The unique identifier of each user.
- `STATUS`: The raw text of the status update.
- `sOPN`: The raw score for Openness.
- `sEXT`: The raw score for Extraversion.
- `sAGR`: The raw score for Agreeableness.
- `sCON`: The raw score for Conscientiousness.
- `sNEU`: The raw score for Neuroticism.

The profiling report confirmed that there are zero missing cells in the dataset. However, 16 duplicate rows were identified and subsequently removed to prevent duplicate data points in our embeddings.

Additionally, we observed that the Neuroticism variable (`sNEU`) is negatively correlated with all other personality traits. Notably, it displays a strong negative correlation with Extraversion (-0.419), as illustrated in Figure 1.

3 Data Pre-Processing and Embedding

3.1 Data Cleaning and Pre-Processing

To ensure a uniform textual structure across status updates before generating vector representations, we implemented a custom pre-processing function that involved case folding, filtering for English-only posts via `langdetect` (retaining 8,594 rows) to ensure the clustering algorithm groups posts based on semantic themes rather than language differences, converting emojis to text, removing noise (URLs, HTML, and mentions), and stripping non-essential punctuation.

3.2 Standardizing Personality Scores

In social data, raw personality scores often exhibit baseline scale bias (for instance, people tend to rate themselves higher in Openness than in Neuroticism). Comparing raw averages directly across different traits within a cluster can lead to misleading conclusions.

To make the traits directly comparable, we utilized scikit-learn's `StandardScaler` to standardize the scores across all unique users. Under this standardized scale, we accurately identify which specific traits are uniquely elevated or suppressed within each cluster.

3.3 Embedding

Once the textual captions were processed, we generated dense vector embeddings using the `all-MiniLM-L6-v2` model from the Sentence-Transformers framework. The model was selected for its optimal balance between computational efficiency and semantic accuracy.

Despite its compact 6-layer, 384-dimensional architecture, it is trained on over one billion sentence pairs specifically to group semantically similar texts closely together. This smaller vector size ensures rapid processing speeds and provides a more stable baseline for dimensionality reduction (UMAP) and density clustering (HDBSCAN).

4 UMAP and HDBSCAN Clustering

To partition the generated SBERT embeddings, we utilize a combined UMAP and HDBSCAN pipeline. This approach is similar to the methodology of Bushmelev et al. (2024)[1], who successfully mapped semantic behavioral clusters to Big Five personality profiles.

4.1 UMAP

To mitigate the distance-metric dilution of high-dimensional spaces, we apply Uniform Manifold Approximation and Projection (UMAP). UMAP reduces the 384-dimensional embeddings to 5-dimensional space for clustering, preserving local and global semantic topologies, and subsequently to 2 dimensions for Cartesian visualization.

4.2 HDBSCAN

We perform clustering using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). HDBSCAN is ideal for social media text because it discovers the optimal number of clusters natively, handles varying conversational densities, and automatically isolates unstructured updates as noise (labeled as -1).

To identify cohesive clusters, we executed a grid search across key parameters: min_cluster_size, min_samples, cluster_selection_epsilon, and cluster_selection_method, evaluating configurations via the Silhouette Score and Davies-Bouldin Index. The optimal configuration yielded 32 distinct clusters with a noise level of 43.80% (Silhouette: 0.5533; Davies-Bouldin: 0.5575), using a min_cluster_size of 30, min_samples of 25, cluster_selection_epsilon of 0.0, and the 'eom' (Excess of Mass) selection method.

4.3 Semantic Keyword Extraction

Finally, we filtered out standard English stopwords and conversational noise (such as "y'all" and "just"), then applied a cluster-level Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer. Treating each cluster's combined status updates as a single document, TF-IDF isolates and extracts the most representative keywords for each cluster.

5 Results

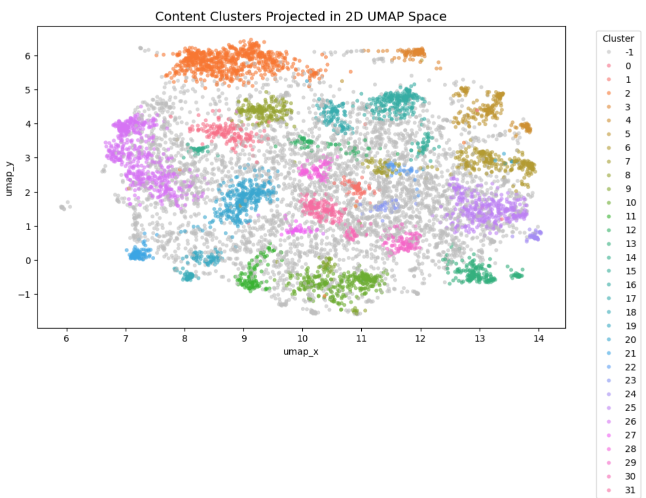

Figure 2: Spatial distribution of the 32 identified semantic clusters in 2D UMAP space.

Ultimately, our pipeline identified 32 distinct linguistic clusters alongside a noise cluster, as shown in Figure 2. In the visualization, each color-coded region represents a unique semantic cluster, while the light grey points (designated as Cluster -1) represent the noise points filtered out by the HDBSCAN algorithm, which contains 3,763 status updates.

To analyze the personality profiles of the linguistic cohorts, we calculated the standardized mean scores of each personality trait within each cluster. These values were then directly compared against the standardized global population mean.

To evaluate the semantic and psychometric properties of each cluster, we examined the distinct keywords that define each group, the cluster size, and the most and least dominant standardized personality traits. The complete psychometric profiles of these 32 clusters are summarized in Table 1.

We selected several clusters to discuss, that seemed to exhibit behaviors contrasting conventional expectations:

- **Cluster 11 (*writing, paper, essay, book*):** Although academic tasks typically associate with high Conscientiousness (C), this cohort displays high Agreeableness (A, $+0.308$ SD) and low Conscientiousness (C, -0.241 SD). This indicates that instead of quiet studying, these users utilize social media to vent about academic pressure and seek peer solidarity. Posting about looming deadlines serves as an interpersonal bonding mechanism to establish shared relatability, driving up Agreeableness at the expense of self-reported discipline.

- **Cluster 4 (*year, resolutions, decade, 2010, 2012*):** This cluster exhibits elevated Neuroticism (N, +0.153 SD) and low Conscientiousness (C, −0.252 SD). The high Neuroticism likely stems from the anxiety of goal-setting and major life transitions, alongside the era's cultural fatalism surrounding the predicted "2012 apocalypse." The low Conscientiousness aligns with the psychology of resolutions: highly disciplined individuals maintain quiet, consistent routines year-round, whereas those lower in self-discipline are more prone to publicly broadcasting temporary, highly aspirational goals.
- **Cluster 12 (*hair, tattoo, blonde, haircut*):** Rather than conventional grooming, this alternative aesthetic cluster displays high Neuroticism (N, +0.289 SD) and low Agreeableness (A, −0.287 SD). Psychologically, sudden physical modifications (like haircuts or tattoos) can act as a coping mechanisms for emotional volatility, aligning with Neuroticism. Concurrently, low Agreeableness reflects nonconformity and individualism, indicating a desire to assert a distinct personal identity that intentionally challenges standard societal expectations.
- **Cluster 8 (*movie, watching, watched, avatar*):** This cluster exhibits elevated Neuroticism (N, +0.252 SD) and suppressed Agreeableness (A, −0.088 SD), indicating highly critical and emotive media consumption. Elevated Neuroticism points to strong emotional sensitivity toward upsetting storylines and plot twists. Meanwhile, the lower Agreeableness reflects the argumentative nature of online film debates, where users actively challenge character decisions, and debate conflicting interpretations of how a movie should have ended.

Cluster	Representative Keywords	Size	Dominant Trait	Least Dominant Trait
0	song, music, album, band, songs	162	O (+0.156)	C (-0.109)
1	gym, workout, yoga, muscles, body	73	C (+0.200)	O (-0.267)
2	wedding, movie, married, coming, awesome	694	E (+0.173)	C (-0.079)
3	birthday, wishes, thanks, special, lucky	89	E (+0.321)	N (-0.270)
4	year, resolutions, decade, 2010, 2012	46	N (+0.153)	C (-0.252)
5	christmas, merry, thanksgiving, santa, family	164	A (+0.066)	O (-0.210)
6	snow, rain, cold, weather, winter	247	A (+0.182)	N (-0.064)
7	car, driving, insurance, license, driver	59	E (+0.219)	C (-0.201)
8	movie, watching, movies, avatar, watched	198	N (+0.252)	A (-0.088)
9	job, interview, resume, mba, jobs	47	E (+0.290)	O (-0.245)
10	class, school, exam, study, semester	303	E (+0.021)	O (-0.086)
11	writing, paper, essay, book, reading	109	A (+0.308)	C (-0.241)
12	hair, black, tattoo, blonde, haircut	66	N (+0.289)	A (-0.287)
13	football, steelers, game, basketball, baseball	182	E (+0.265)	O (-0.221)
14	dream, dreams, dreaming, reality, meaning	34	O (+0.420)	N (-0.177)
15	hangover, beer, drunk, drink, alcohol	45	E (+0.006)	C (-0.374)
16	cheese, chocolate, soup, cake, food	231	A (+0.256)	E (-0.151)
17	zoo, animal, bunny, dog, animals	96	N (+0.042)	E (-0.172)
18	phone, iphone, google, blackberry, droid	53	A (+0.085)	C (-0.154)
19	computer, internet, laptop, drive, pc	59	N (+0.132)	E (-0.189)
20	thinks, wishes, needs, wants, apartment	328	O (+0.334)	N (+0.022)
21	facebook, click, twitter, status, feed	100	N (+0.168)	C (-0.253)
22	cleaning, laundry, packing, washed, clean	36	E (+0.330)	N (-0.585)
23	wonderful, church, evening, event, nice	32	E (+0.416)	N (-0.206)
24	comedy, downtown, club, jokes, hosting	40	E (+0.230)	A (-0.792)
25	chicago, trip, surf, flight, beach	360	E (+0.158)	N (-0.196)
26	god, lord, world, heart, mind	549	O (+0.422)	N (-0.174)
27	bored, tunes, talk, ambition, mystery	40	N (+0.162)	O (-0.181)
28	sick, flu, cold, fever, vaccine	83	E (+0.218)	O (-0.166)
29	friday, monday, sunday, saturday, thursday	107	E (+0.297)	N (-0.279)
30	working, busy, job, relax, afternoon	39	A (+0.196)	N (-0.334)
31	tired, awake, bed, asleep, insomnia	158	N (+0.168)	C (-0.410)

Table 1: Summary of the 32 clusters, their representative keywords, sizes, and strongest/weakest personality traits (expressed as Standardized SD deviations).

6 Conclusions

To address the first part of our research question, our pipeline successfully partitioned 56.2% of the dataset into 32 cohesive semantic clusters, isolating the remaining 43.8% as noise. This confirms that spontaneous social media language is not merely chaotic but is organized around distinct situational, and emotional categories that can be reliably identified using SBERT, UMAP, and HDBSCAN.

To address the second part of our research question, standardizing these clusters against an unbiased unique-user baseline revealed highly distinct, often counter-intuitive Big Five profiles. Rather than aligning with conventional stereotypes, several clusters demonstrated complex psychological signaling. This includes academic procrastination venting mapping to low Conscientiousness but high Agreeableness (Cluster 11), and new year discussions mapping to elevated Neuroticism (Cluster 4) due to existential anxiety and the cultural fatalism of the predicted “2012 apocalypse.” Ultimately, these findings confirm that spontaneous online language can serve as a viable medium for studying personality, revealing complex emotional dynamics that conventional self-report questionnaires often overlook.

References

[1] Fedor Bushmelev, Valeriia Stoliarova, and Tatiana Tulupyeva. Semantic based clusters of vk users avatars and their association with the big five personality profiles. In *Proceedings of the Eighth International Scientific Conference "Intelligent Information Technologies for Industry" (IITI'24), Volume 2*, pages 183–192, 2024.

[2] Sridhar Srinivasan. Mypersonality_nlp. https://github.com/sridharsrinivasan7/MyPersonality_NLP, 2024.